

CODE

```
import math
```

```
Blasting_Diameter = float(input("Enter Blasting Diameter (mm): "))
```

```
Average_In_Hole_Density = float(input("Enter Average In-Hole Density (kg/m3): "))
```

```
Spacing_To_Burden_ratio = float(input("Enter Spacing:Burden Ratio: "))
```

```
Bench_Height = float(input("Enter Bench Height (m): "))
```

```
Proximity = input("Enter 'Close to Community' or 'Not': ").upper()
```

```
Rock_Type = input("Enter Rock Type (Hard/Soft): ").upper()
```

```
Technical_Powder_Factor = float(input("Enter your Technical Powder Factor: "))
```

```
Mass_of_an_explosive = (Average_In_Hole_Density * Blasting_Diameter**2) / 1273
```

```
if Proximity == "CLOSE TO COMMUNITY":
```

```
    Stemming_Length = 30 * Blasting_Diameter
```

```
    Burden = math.sqrt((Mass_of_an_explosive * (Bench_Height - Stemming_Length)) /  
(Spacing_To_Burden_ratio * Technical_Powder_Factor))
```

```
elif Proximity == "NOT":
```

```
    Stemming_Length = 20 * Blasting_Diameter
```

```
    Burden = math.sqrt((Mass_of_an_explosive * (Bench_Height - Stemming_Length)) /  
(Spacing_To_Burden_ratio * Technical_Powder_Factor))
```

```
else:
```

```
    print("Invalid Proximity.")
```

```
Spacing = Spacing_To_Burden_ratio * Burden
```

```
subdrill = Burden / 3
```

```
if Rock_Type == "HARD":
```

```
    Actual_Powder_factor = (Bench_Height - Stemming_Length + subdrill) / (Burden *  
    Spacing * Bench_Height)
```

```
elif Rock_Type == "SOFT":
```

```
    Actual_Powder_factor = (Bench_Height - Stemming_Length - subdrill) / (Burden *  
    Spacing * Bench_Height)
```

```
else:
```

```
    print("Invalid Rock Type.")
```

```
print("=" * 10, "OUTPUT", "=" * 10)
```

```
print("Burden =", Burden, "m")
```

```
print("Spacing =", Spacing, "m")
```

```
print("Stemming Length =", Stemming_Length, "m")
```

```
print("Actual Powder Factor =", Actual_Powder_factor)
```

```
print("=" * 10, "END", "=" * 10)
```

```
Next_Step = input("Want to find MeanFragSize (YES/NO): ").upper()
```

```
if Next_Step == "NO":
```

```
    print("Thank You!")
```

```
elif Next_Step == "YES":
```

```
    Bench_Height = float(input("Enter your BenchHeight: "))
```

```
    Relative_Effective_Energy_of_Explosive = float(input("Enter your Relative effective  
    energy of explosive: "))
```

```
    Stemming_Length_2 = float(input("Enter your Stemming Length: "))
```

```
Mean_Fragment_Size = (Average_In_Hole_Density*(Bench_Height-  
Stemming_Length_2))/(Relative_Effective_Energy_of_Explosive)
```

```
print("Mean Fragment Size=", Mean_Fragment_Size,"mm")
```

```
else:
```

```
print("Invalid!")
```